

Implementation and Defense Against Adversarial Attacks in Image Classification Model

BY

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Dissertation

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ABSTRACT

Adversarial attacks exploit the vulnerabilities of machine learning models, particularly deep learning models like CNNs, by introducing carefully crafted perturbations to the input data. These perturbations are usually small enough to be imperceptible to the human eye but can cause the model to misclassify the input. For example, an image of a panda might be subtly altered so that a CNN model mistakenly identifies it as a gibbon, while to a human observer, the image still clearly appears to be a panda. The primary goal of these attacks is to fool the model into making incorrect predictions, either by changing the model's output to a specific incorrect label (targeted attack) or simply causing any incorrect prediction (untargeted attack). This can have serious implications, especially in sensitive applications like autonomous driving, medical diagnosis, or security surveillance. CNN models learn to recognize patterns in images by using a series of convolutional filters and non-linear transformations. However, they can be overly sensitive to small, seemingly random changes in the input data. Adversarial attacks exploit this by adding a calculated "noise" to the input image that shifts the model's decision boundary.

However various defense techniques such as Adversarial Training, usage of Defense Layer (DefendedModel) and image augmentation like Noise Injection Techniques can be used to promote robustness against such attacks.

Keywords:

Adversarial Attacks

Perturbations

Noise

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Chapter 1

Introduction

1.1 Background

Deep learning has significantly impacted various sectors including autonomous vehicles, healthcare systems, biometric identification, and surveillance technologies. Among the most effective tools in this space are Convolutional Neural Networks (CNNs), which have demonstrated high accuracy and strong generalization capabilities in handling complex image-based tasks. Nevertheless, these models are highly susceptible to adversarial attacks—small, intentional perturbations introduced to input data to mislead the model, often without any noticeable change to the human eye. One of the earliest forms of adversarial attack appears through the implementation of Fast Gradient Sign Method (FGSM) [1]. The procedure determines minor data modifications according to loss function gradients which makes the network mistake its data classification decisions. The simplicity and effectiveness of FGSM highlight the inherent weaknesses of deep learning systems, especially when applied in sensitive areas like medical imaging, facial authentication, or self-driving technology, where prediction errors can have serious implications.

As adversarial machine learning has matured, research has expanded to include both advanced attack strategies and a variety of defense mechanisms. Defensive measures include methods like adversarial training—where models are retrained using adversarial inputs—and input preprocessing techniques such as smoothing, denoising, or compression. However, many of these defenses face limitations: they may reduce accuracy on unperturbed inputs, require substantial computational power, or lack effectiveness against a broad spectrum of attacks.

1.2 Motivation

The crucial essence of deep learning model reliability and accuracy continues to grow in significance with the increasing adoption of artificial intelligence in practical applications. The discover of how data input alterations affect deep learning models negatively impacts their suitability for security-critical situations. During autonomous vehicle operation an incorrect identification of traffic signs can produce severe accidents. Incorrect image classifications within medical scenarios could cause the misdiagnosis along with improper treatment methods.

The FGSM emerges as an attention-grabbing adversarial technique because it offers high efficiency alongside straightforward implementation. The simple nature of Fast Gradient Sign Method (FGSM) lets researchers and attackers use it to examine neural network vulnerabilities while requiring low computational resources.

The defensive measure of adversarial training enables models to improve their resistance through the use of perturbed inputs during retraining processes. Understanding and manipulating incoming data through preprocessing strategies represents an effective method for removing potential attacks. During training models become more robust when noise inputs including Gaussian noise along with image transformation techniques are integrated into the process for broad representation learning.

Such evaluation combines an analysis with a direct comparison between these defensive strategies. The objective seeks to discover efficient pairs of protection mechanisms that defend against FGSM attacks to enhance deep learning security.

1.3 Proposal

This thesis proposes a systematic study of FGSM adversarial attacks and multiple corresponding defense mechanisms to enhance model robustness. Specifically, it focuses on evaluating how different deep learning models both complex and lightweight respond to adversarial attacks and what strategies can be implemented to mitigate these effects.

The problem addressed is two fold:

1. **Evaluating Vulnerability:** Understanding the impact of FGSM on CNNs, including high-capacity pretrained models like *ResNet-34* and manually con-

structed models like *myCNN*, when applied to datasets such as Kaggle’s image classification dataset and *MNIST*.

2. **Implimenting and Testing Defenses:** Implementing and analyzing various defense techniques, including:

- **Adversarial Training:** Retraining the model using a mix of clean and FGSM-perturbed images to build resilience.
- **Defense Layer (DefendedModel):** Introducing preprocessing steps like *Gaussian blur*, *JPEG compression*, and *median denoising* to clean adversarial inputs before classification.
- **Noise Injection Techniques:** Training models on augmented data with injected noise patterns such as *Random Gaussian Noise (RGN)*, *Stochastic Motion Blur (SMB)*, *Stochastic Glass Blur (SGB)*, and *Random Sized Coarse Dropout (RSCD)* to promote robustness.
- **Multiple Epsilons Training:** This involves generating adversarial examples from the training dataset and including them in the training process. This helps the model learn more resilient decision boundaries rather than using a single, fixed perturbation strength ϵ , adversarial samples were generated using multiple epsilon values: $\epsilon \in \{0.01, 0.05, 0.1, 0.2, 0.3\}$.

By comparing these approaches under different epsilon levels (perturbation strengths), the thesis aims to identify which techniques or combinations provide the most reliable defense against FGSM, without severely compromising accuracy on clean data. The main goal targets the development of deep learning systems which maintain security and resilience for trusted deployment in real-world operations.

1.4 Contributions

This thesis delivers the following main contributions:

1. **Comprehensive Evaluation of FGSM Attacks:**

The thesis section performs an organized assessment of FGSM adversarial attacks on ResNet-34 complex models combined with *myCNN* lightweight models using the Kaggle Image Classification dataset and *MNIST*.

2. Implementation of Adversarial Training and Epsilon Training:

Both models receive adversarial training as a defensive approach which enhances their ability to withstand FGSM attacks with an evaluation of the clean accuracy versus robustness relationship.

3. Designing Defense Layer Architecture:

A novel DefendedModel (defModel) architecture is proposed, which integrates image preprocessing techniques such as Gaussian blur, JPEG compression, and median denoising to suppress adversarial perturbations before classification.

4. Investigation of Noise Injection Techniques:

The part of thesis explores the effectiveness of noise injection as a passive defense strategy. Four different augmentation-based defenses — *Random Gaussian Noise (RGN)*, *Stochastic Motion Blur (SMB)*, *Stochastic Glass Blur (SGB)*, and *Random Sized Coarse Dropout (RSCD)* — are tested for their impact on adversarial robustness.

5. Experimental Analysis across Multiple Defense Strategies:

A comparative study is conducted on all implemented defenses across various epsilon values, highlighting the strengths and limitations of each method under different levels of adversarial intensity.

Chapter 2

Literature Survey

Throughout the years, deep learning models have attracted significant attention due to their susceptibility to adversarial attacks, which can mislead even the most stable architectures. The research by [1] explained the concept of adversarial attacked examples and emphasized the necessity for more robust models to counter such threats. Their work, presented at the ICLR, Revealed built-in flaws present in deep learning models. Similarly, [2] introduced DeepFool, a good and effective method for generating adversarial examples that achieved high success rates in deceiving classifiers while maintaining low computational complexity.

Building on these foundational works, [3] proposed adversarial training, a defense technique aimed at enhancing model resilience by incorporating adversarial examples during the training process. Also presented at ICLR, their approach demonstrated how carefully designed training procedures could improve resistance to perturbations. However, detecting adversarial examples remains a formidable challenge, as shown by [4], who revealed that many detection frameworks can be bypassed with specially crafted adversarial inputs.

Further advancements in defense strategies have been discussed by Chen et al. [5], who offered a practical perspective on attacks and defenses in image classification. Their study highlights the dual necessity of theoretical rigor and practical applicability in building robust models. In a related effort, [6] proposed FENCEBOX, a platform employing data augmentation techniques as a primary line of defense against adversarial attacks, which demonstrated strong potential in reinforcing model robustness.

A significant portion of defensive research focuses on data augmentation. The research [7] conducted a comprehensive review of image augmentation methods—including

rotation, flipping, and color jittering—to enhance the generalization and robustness of deep learning systems. Similarly, [8] examined diverse preprocessing and augmentation approaches, providing valuable perspectives on their role in countering adversarial attacks.

To address adversarial robustness from another angle, some researchers have turned to compression-based defense mechanisms. [9] introduced Feature Distillation, a method combining DNN-specific JPEG compression with defense strategies to reduce the effect of adversarial noise. Concurrently, [10] introduced a method for variable-rate image compression leveraging recurrent neural networks, presenting a potential strategy for mitigating adversarial effects by compressing key image features.

Lastly, the issue of adversarial defense continues to be an active area of exploration. Research [11] investigated how adversarial training contributes to model stability and resilience, presenting it as a robust technique to improve performance under adversarial conditions.

Chapter 3

Preliminaries

Adversarial attacks exploit vulnerabilities in machine learning models, particularly deep learning frameworks like Convolutional Neural Networks (CNNs). By subtly altering input data in ways that are visually imperceptible yet significantly affect model performance. These modifications, though minimal and often not visible to human eye, are enough to cause the model to produce incorrect results. For instance, an image of a panda can be slightly altered in such a way that a CNN incorrectly identifies it as another animal like a gibbon, even though the visual difference is undetectable to humans.

The intent behind these attacks is to mislead the model—either by forcing it to predict a specific incorrect label (in the case of targeted attacks) or any wrong label (in untargeted attacks). Such misclassifications can be highly dangerous in critical and important systems like self-driving vehicles, healthcare or medical diagnostics, or security applications.

CNNs operate by identifying visual features through layered convolutional operations and nonlinear activations. However, they tend to be highly reactive to minor input changes, even if those changes appear insignificant. Adversarial techniques exploit this vulnerability by injecting calculated distortions or "noise" that pushes the input across the model's decision boundaries, leading to incorrect predictions.

3.1 Adversarial Attacks

Over time, various adversarial attack techniques have emerged, all designed to subtly modify input data, such as images, in ways that lead to misclassification by machine

learning models. For instance, how might one alter an image that is correctly classified as a goldfish by a model, such that it now gets misclassified, even though a human would still clearly identify it as a goldfish? The visual content remains unchanged for human perception, but the model’s interpretation is successfully manipulated.

This concept aligns closely with the generator component of Generative Adversarial Networks (GANs), whose purpose is to produce inputs that can mislead another network, known as the discriminator. The ability to deceive these models highlights a larger concern: numerous advanced machine learning systems, including cutting-edge deep learning networks, remain vulnerable to adversarial inputs. These inputs differ only marginally from correctly classified examples but still lead to wrong classification.

What’s particularly concerning is that these adversarial inputs can deceive various models, regardless of their architecture or training datasets. This indicates that the vulnerability is not model-specific but a systemic issue in the learning process itself. Initially, the underlying causes remained uncertain, with initial theories suggesting that the intricate non-linear characteristics of deep networks were to blame. Also the lack of regularization, or insufficient model averaging.

However, later findings suggest that such complexity is not a prerequisite for adversarial vulnerability. Instead, even simple linear characteristics in high-dimensional spaces can explain the emergence of these adversarial examples. This realization has led to more efficient methods for generating adversarial inputs and has made adversarial training—a defense mechanism—more practical.

Moreover, adversarial training adds a layer of regularization beyond what traditional techniques like dropout or pretraining offer. Interestingly, generic regularization techniques have limited impact on reducing adversarial risk, while switching to fundamentally different architectures—like radial basis function (RBF) networks—can improve robustness. This highlights a tradeoff between ease of training with linear models and increased resistance to attacks from more nonlinear models. Overcoming this challenge may depend on future advancements in optimization techniques that can effectively train complex, nonlinear models.

3.1.1 Linear Perspective on Adversarial Examples

To understand the emergence of adversarial examples, we begin by considering linear models. In many real-world tasks, particularly those involving digital media such as images, the numerical precision of input features is inherently limited. For example, pixel intensities in 8-bit images can only represent 256 discrete levels, meaning changes below $\frac{1}{255}$ are effectively ignored.

Due to this quantization, a classifier should ideally behave consistently when presented with two inputs that differ by less than this minimum detectable change. More formally, given an original input x and a slightly perturbed or modified input $\tilde{x} = x + \eta$, the model should ideally predict the same class if the perturbation satisfies:

$$\|\eta\|_{\infty} < \epsilon, \quad (3.1)$$

where ϵ is a small threshold that represents the sensitivity limit of the input encoding process.

Now, let us evaluate how a linear model reacts to such perturbations. For a linear model with a weight vector w , the activation for the perturbed input $\tilde{x}$ becomes:

$$w^{\top} \tilde{x} = w^{\top} x + w^{\top} \eta. \quad (3.2)$$

The term $w^{\top} \eta$ represents the impact of the perturbation on activation of CNN model. To maximize this effect while keeping η within the allowed limit, the optimal choice is:

$$\eta = \epsilon \cdot \text{sign}(w), \quad (3.3)$$

which aligns the perturbation with the direction of the weights. If the vector w has n components and the average absolute value of each component is m , then the overall increase in activation becomes:

$$\epsilon \cdot m \cdot n. \quad (3.4)$$

Interestingly, while the individual perturbation values remain minuscule, their combined influence on the output grows proportionally with the number of input dimensions n . This implies that many imperceptible changes can accumulate into a significant effect on the prediction.

This phenomenon can be seen as a kind of unintentional information hiding, where the model becomes overly sensitive to the input features most aligned with its weights, even if they are not the most dominant signals in the input.

In conclusion, even linear models are susceptible to adversarial examples when the input dimensionality is high. This observation provides a simpler and more fundamental explanation than prior hypotheses that focused on the complex, non-linear characteristics of neural networks. It also clarifies why even softmax regression, a basic linear classifier, can give wrong results on adversarial perturbations.

3.2 FGSM

The Fast Gradient Sign Method (FGSM) established by [1] serves as a basic yet widely recognized method to produce adversarial examples. FGSM produces adversarial perturbations through mathematical derivatives of neural network loss function gradients applied to input images. The modified image results from this method looks practically identical to the original version to human observers but leads to significant changes in network prediction output.

The FGSM algorithm computes the loss function (cross-entropy) gradient regarding the input image before adding a gradient sign scale to the original input. The mathematical representation of FGSM is provided in Equation 4.1:

$$outputImage_{adv} = inputImage + \epsilon \cdot \text{sign}(\nabla_x L(\theta, inputImage, trueLabel)) \quad (3.5)$$

Where:

- $outputImage_{adv}$ denotes the adversarially altered image,
- $inputImage$ is the original input image,
- $trueLabel$ represents the true label of the input,
- ϵ is constant value that controls the perturbation size,
- θ indicates the model's parameters,
- L is the loss function used in model training,

- $\nabla_x L(\theta, inputImage, trueLabel)$ is the gradient of loss function with respect to the input image.

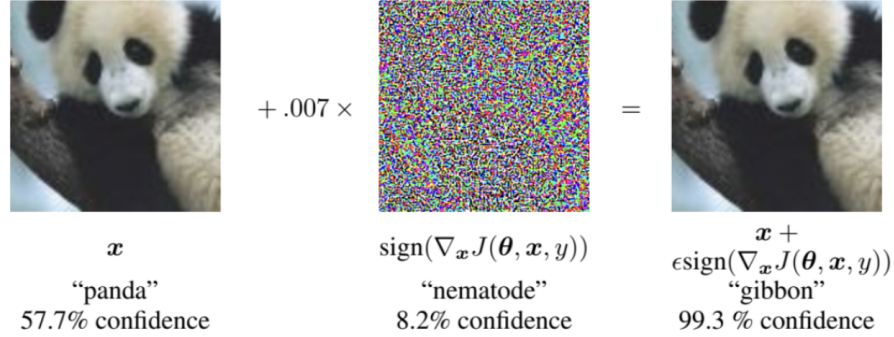


Figure 3.1: Example of FGSM attack. Adding a small noise or perturbation with $\epsilon = 0.007$ corresponding to least significant bit in 8-bit image encoding successfully causes misclassification, while remaining imperceptible to the human eye. Adapted from [1].

3.2.1 Procedure of FGSM Attack

The implementation of the FGSM attack involves the following three primary steps:

1. **Forward Computation:** The input image passes through the model for prediction generation and the loss operation utilizes an appropriate objective function for measurement.
2. **Gradient Calculation:** Calculate the gradient of the loss function with respect to the input image pixels.
3. **Generating Perturbation:** Alter the input image by adding a small, directed noise based on the sign of the gradient, aiming to mislead the model into incorrect classification.

3.2.2 Understanding Gradient Manipulation

In standard neural network training, Gradients guide the adjustment of model parameters (weights) in a way that reduces the value of the loss function. This process allows the model to gradually improve its predictions over time through backpropagation.

The core idea is to use these gradients intentionally to increase the loss function. By modifying the input image in the direction that maximizes loss function, FGSM generates adversarial examples that cause the model to make incorrect predictions. Notably, these input noise are so small that they remain invisible to the human eye, yet they significantly influence the model’s decision.

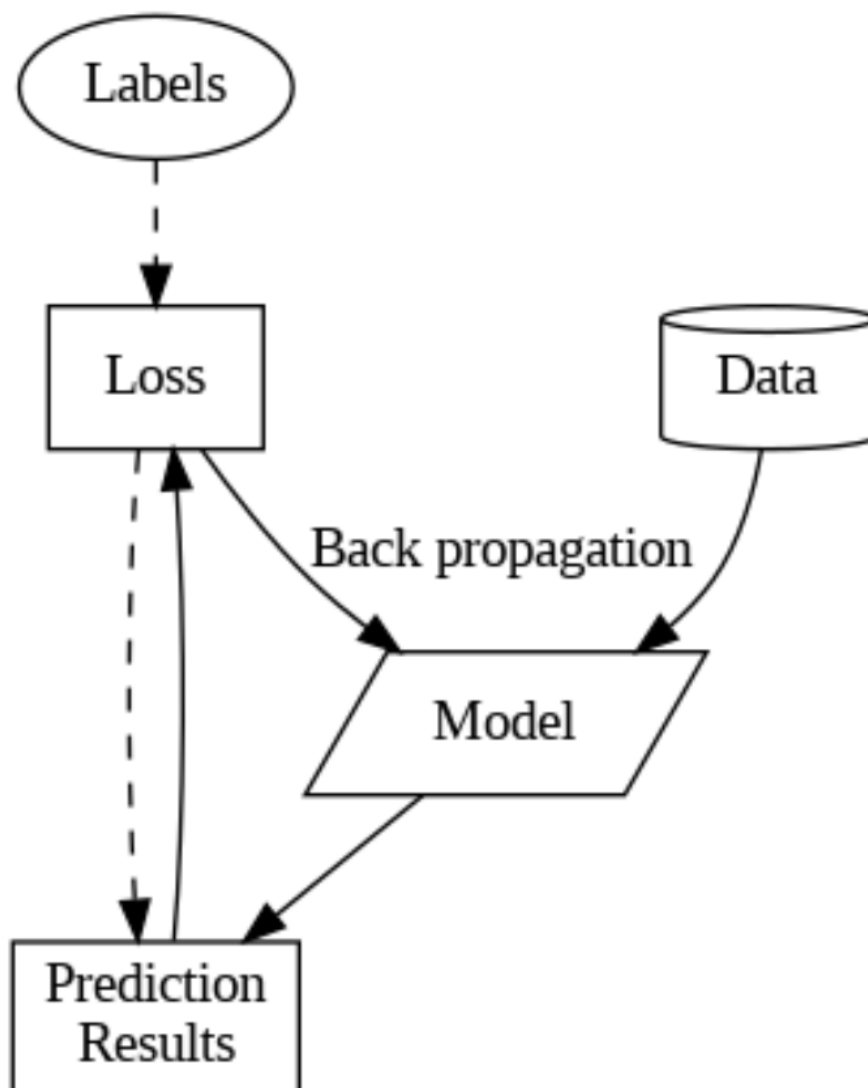


Figure 3.2: Conventional training of neural network model.

Figure 3.2 illustrates the conventional training process, where models learn from clean input data by minimizing prediction errors via backpropagation. This process helps the model generalize to unseen data.

Chapter 4

Proposed Work

4.1 Adversarial Defense Using Adversarial Training

Machine learning models get their resistance to adversarial inputs primarily through the implementation of adversarial training methods. Adversarial training replicates the process of training with basin comprising labeled data and perilous data samples which allows the model to acquire resilient features against harmful influences. Although it succeeds in making models more robust, adversarial training comes with some challenges. They are, for instance, higher computational overhead, higher memory usage, and restrictions on the level of diversity in the created adversarial examples, which can have a harmful impact on the model's generalization ability.

Various solutions have been developed to counter these limitations. One significant approach is incorporating adaptive adversarial perturbations into the model parameter space to make training seamless without necessitating huge adversarial datasets. Another method, single-step adversarial training with dropout scheduling, has proven to be effective in protecting against fast gradient-based attacks at the cost of low training expenses. Likewise, Multi-Stage Optimization-Based Adversarial Training seeks to reconcile robustness and computational simplicity and minimize overfitting due to repeated adversarial exposure.

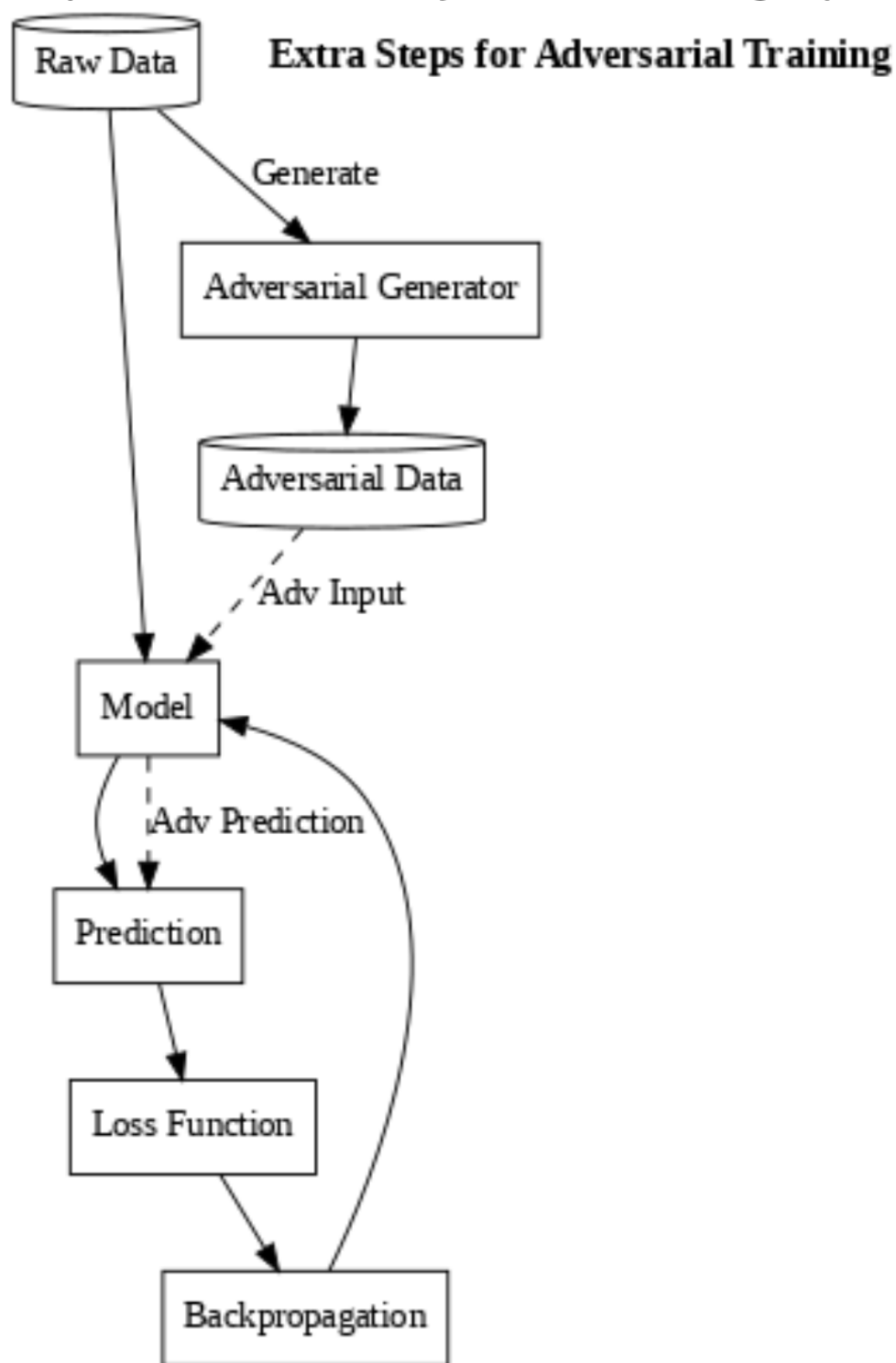


Figure 4.1: Adversarial training of neural network model.

Figure 4.1 depicts training on adversarial examples, where an additional step is included to generate adversarial samples during training. This inclusion forces the model to learn from both clean and perturbed data, thereby enhancing its defense mechanisms. Beyond these methods, lightweight defense strategies such as label smoothing and logit squeezing have been introduced. These techniques serve as regularization methods that mimic the effect of adversarial training without explicitly generating adversarial samples. Additionally, Adversarial Training with Transferable Adversarial Examples improves training efficiency by leveraging adversarial examples generated across different epochs to enrich perturbation diversity.

The application of adversarial training has proven beneficial toward improving safety standards in essential systems involving medical diagnosis and autonomous vehicles. For example, by applying adversarial training to the vision system of an autonomous car, manufacturers can enhance its robustness to subtle adversarial distortions, ensuring safer navigation in complex environments. Recent studies on FGSM-based adversarial training focus on boosting robustness, reducing training time, and addressing overfitting issues. These advancements continue to drive the evolution of adversarial defense techniques, making neural networks more secure and reliable for deployment in high-risk domains.

4.1.1 Methodology

This work investigates the weakness of deep learning models on adversarial inputs using the FGSM and demonstrates how adversarial training can bolster model robustness. The classification task is conducted on an image dataset, typically organized using PyTorch’s ImageFolder utility. The data is divided into training and test sets and preprocessed using standard transformations such as resizing and normalization to ensure consistency in input dimensions and distribution.

The task uses a CNN model with a basis on the pretrained ResNet-34 architecture for classification. Because of its replacement with a number of classes equal to the dataset size the final fully connected layer enables training for the distinct classification problem. The training algorithm utilizes the Adam optimizer to optimize the model based on the cross-entropy loss function for achieving minimum errors when predicting clean inputs.

After the initial training, the accuracy of the model is examined on a held-out test

set, achieving baseline performance typically around 90–95%. To assess adversarial vulnerability, the FGSM attack is implemented. This method perturbs the input image in the direction that maximally increases the loss, guided by the gradient with respect to the input. The perturbed image is computed as:

$$outputImage_{adv} = inputImage + \epsilon \cdot \text{sign}(\nabla_x L(\theta, inputImage, trueLabel)) \quad (4.1)$$

where *inputImage* is the original image, θ represents the parameters of the model, L is the loss function, and ϵ is a small scalar that controls perturbation strength. Though visually similar to *inputImage*, the adversarial image often causes misclassification by the model.

When these adversarial examples are passed through the trained network, a significant drop in accuracy is observed—for instance, from 95% on clean images to around 60% on perturbed images—highlighting the model’s sensitivity to even minimal input changes.

To counter this issue, adversarial training is applied. In this procedure, each mini-batch during training contains both clean and adversarial images. For every clean sample, the gradient is computed and used to generate an adversarial version. The model is then updated using the combined set of clean and adversarial inputs, encouraging robustness across both domains.

Post adversarial training, the model shows significantly improved resistance to input perturbations. While clean image accuracy may slightly decrease (e.g., from 95% to 91%), adversarial accuracy improves markedly (e.g., from 63% to 84%). This trade-off reflects the essence of robust learning, where increased security against adversarial manipulation is achieved at the cost of a minor drop in clean performance.

Overall, this methodology validates the FGSM attack’s effectiveness in exposing model weaknesses and underscores the importance of adversarial training in constructing models suitable for deployment in critical environments such as autonomous vehicles and medical or clinical diagnostics.

Algorithm 1 Standard FGSM Adversarial Training for T Epochs

Input: Perturbation scale ϵ , Number of epochs T , Dataset size M **Output:** Updated model parameters θ **for** $t = 1$ to T **do** **for** $i = 1$ to M **do** Compute gradient: $g \leftarrow \nabla_x \ell(f_\theta(x_i), y_i)$ Generate adversarial input: $x_i^{\text{adv}} \leftarrow x_i + \epsilon \cdot \text{sign}(g)$ Update model: $\theta \leftarrow \theta - \nabla_\theta \ell(f_\theta(x_i^{\text{adv}}), y_i)$ **end for****end for**

This algorithm strengthens models against attacks by training them on intentionally perturbed examples, forcing them to learn robust features that work even with small input modifications.

4.2 Adversarial Defense Using Noise Injection

Adversarial attacks are the significant threat to the robustness and reliability of neural networks. In this experiment, we explore the use of noise injection as a defense mechanism to counter adversarial inputs or examples generated using the FGSM. We use a modified ResNet-34 architecture trained on the MNIST dataset and examine how various noise techniques—applied to adversarial inputs. It can enhance robustness of the model and resistance against attacks.

4.2.1 Methodology

4.2.1.1 Dataset and Preprocessing

This experiment uses the commonly acknowledged MNIST dataset including 70,000 grayscale handwritten digit images (0 through 9) with a 28×28 pixel resolution. The analysis requires the selection of 2000 training images while using all test images for evaluation.

Each image is normalized by dividing pixel intensities by 255 and is expanded to include a channel dimension to suit CNN input requirements. Since the ResNet model expects 3-channel RGB inputs, all grayscale images are converted to RGB by

repeating the grayscale channel three times. The images undergo resizing to become 32×32 pixels to fulfill the input requirements of ResNet-34 architecture.

4.2.1.2 Model Architecture

For this study researchers apply the ResNet-34 model structure. The model enters training without its weights set from previous experiences and its final classification section omitted. The flattened output of the ResNet base moves through a network with 1000 ReLU activated dense units. The system includes a final Dense layer which contains 10 neurons along with softmax activation since MNIST has ten digit classes.

The proposed model receives optimization from the Adam optimizer and performs evaluation through accuracy while using categorical cross-entropy loss. It is trained for 5 epochs on the preprocessed MNIST training data with a validation split of 10%.

Algorithm 2 Training on Adversarial examples with Fixed Noise for T Epochs

Input: Dataset (x_i, y_i) , perturbation ϵ , model f_θ
Output: Updated model parameters θ
for $t = 1$ to T **do**
 for $i = 1$ to M **do**
 // Generate adversarial example using FGSM
 $g \leftarrow \nabla_x \ell(f_\theta(x_i), y_i)$
 $x_{\text{adv}} \leftarrow x_i + \epsilon \cdot \text{sign}(g)$
 // Apply fixed noise sequence
 $x_{\text{noisy}} \leftarrow \text{Dropout}(\text{GlassBlur}(\text{MotionBlur}(\text{Gaussian}(x_{\text{adv}}))))$
 // Update model
 $\theta \leftarrow \theta - \nabla_\theta \ell(f_\theta(x_{\text{noisy}}), y_i)$
 end for
end for

This enhanced algorithm first creates adversarial examples using FGSM, then applies a fixed sequence of noise transformations (Gaussian, MotionBlur, GlassBlur, Dropout) before updating the model parameters, making it robust against both adversarial attacks and common image corruptions.

4.2.1.3 Noise Injection Techniques

Adversarial attacks often exploit specific gradients or patterns in the input space that are not visible to the human eyes but highly effective at deceiving deep learning

models. To counteract this, noise injection techniques can be applied to adversarial samples. These methods aim to disrupt or obscure the attack signal before the input reaches the model. The following four techniques were used in this study:

Random Gaussian Noise (RGN) Random Gaussian Noise (RGN) is a common image augmentation technique. In this Zero-mean Gaussian distribution noise is applied to each pixel throughout image domains by the added mechanism.. It simulates sensor noise or background variations and is often used to improve model generalization.

Adversarial perturbations are typically subtle and precise. Introducing random noise disrupts the gradient-based modifications introduced by attacks such as FGSM, making them less effective. RGN acts as a stochastic mask that conceals adversarial structures.

In this experiment, the standard deviation of the noise is not fixed but randomly selected for each pixel within a range (0.0005 to 0.005), introducing spatial variability. The noisy image is clipped to remain in the valid pixel range.

Stochastic Motion Blur (SMB) Motion blur simulates the effect of movement during image capture, causing pixel values to smear along a particular direction. SMB introduces this blur to reduce sharpness in the image.

FGSM perturbations often depend on precise, high-frequency pixel changes. By blurring these features, SMB reduces the visual sharpness of adversarial noise, weakening its effectiveness.

The technique uses a vertical blur kernel applied using OpenCV's `filter2D` method. The blur is applied channel-wise to maintain RGB consistency.

Stochastic Glass Blur (SGB) Glass blur simulates viewing an image through a textured or frosted surface. It causes random scattering of pixel values, distorting the spatial layout of the image.

SGB modifies spatial relationships, breaking the carefully crafted structures of adversarial examples. It acts as a general-purpose spatial transformer that disrupts pixel-level dependencies.

The technique is implemented using a Gaussian filter from SciPy, applied to each image with a fixed sigma.

Random Sized Coarse Dropout (RSCD) Coarse dropout, also called cutout, randomly masks out large regions or patches of the image, simulating occlusion. The restructuring compels the model to use distributed features instead of localized features.

Adversarial perturbations often focus on specific pixel areas. Removing random spatial blocks disrupts these localized patterns, mitigating the attack effect and encouraging the model to generalize better.

A binary mask is created with a fixed dropout probability and applied across the entire image tensor.

4.2.1.4 Training with Noise-Augmented Adversarial Samples and Evaluation

The model receives retraining with original clean samples that have been jointed with noise-injected adversarial samples. The combined training technique reveals clean and defended adversarial examples to the model thus enhancing its ability to recognize examples from both sets.

Combining different noise layers results in incremental accuracy improvements, indicating the complementary nature of these defenses.

4.3 Defense Layers

Adversarial examples—crafted inputs that deceive machine learning models. The challenges posed by personal or situational failures represent critical obstacles in fields that require high sensitivity like healthcare, finance and autonomous driving. To overcome all these challenges without altering the model’s architecture, preprocessing-based defense strategies can be employed. These defense mechanisms, referred to as ”defense layers,” sanitize input images before they are passed into the model for prediction. In this study, three such techniques are assessed.

4.3.1 Using Gaussian Blur

Gaussian blur is a technique that smoothens the image by applying a Gaussian filter, which helps reduce high-frequency noise and sharp patterns. FGSM attacks often generate localized, high-frequency pixel changes that are undetectable to humans

but can confuse neural networks. By applying Gaussian blur, these sharp gradients are diffused, rendering adversarial patterns less effective. A function(kernel) with a typical dimension of 3x3 or 5x5 scans across any image to apply a weight-averaged calculation for neighboring pixels using Gaussian distributions for value replacement.

4.3.2 Using JPEG Compression

JPEG compression works by reducing image quality through the elimination of less critical pixel information, particularly high-frequency details. Since adversarial perturbations usually reside in these high-frequency components, JPEG compression helps to discard these subtle details, effectively neutralizing the attack while maintaining visual quality. In this process, each image is first encoded to JPEG format with moderate compression (e.g., quality = 75) and then decoded back. This simulates typical image transmission scenarios and acts as a low-cost filtering mechanism.

4.3.3 Using Median Filtering (Input Denoising)

Under median filtering a pixel receives its new value as the median calculation of surrounding pixels. This technique is especially useful in removing impulsive noise, such as isolated perturbations, without drastically changing the structure of the image. It proves effective in denoising adversarial inputs. In this study, a 3x3 median filter was applied to each image before making predictions.

All three techniques were evaluated on a set of FGSM adversarial images across varying epsilon values (0.000, 0.007, 0.010, 0.020, 0.030, 0.050, 0.100, 0.200, 0.300). Each defense showed improvements in accuracy, with Gaussian blur and median filtering providing the most robust defenses against stronger attacks (for $\epsilon \geq 0.1$).

Algorithm 3 Defense Layers for FGSM Adversarial Examples

```

1: Input: Test images  $x$ , Labels  $y$ , Trained model  $f_\theta$ , List of perturbation strengths  $\epsilon$ 
2: Output: Accuracy on defended adversarial images
3: for each  $\epsilon$  in  $\{\epsilon_1, \epsilon_2, \dots, \epsilon_n\}$  do
4:   for each  $(x_i, y_i)$  in  $(x, y)$  do
5:     Generate adversarial image:  $x_{\text{adv}} \leftarrow x_i + \epsilon \cdot \text{sign}(\nabla_x J(f_\theta(x_i), y_i))$ 
6:   end for
7:   Apply one of the following defense layers:
8:     Gaussian Blur:  $x_{\text{def}} \leftarrow \text{GaussianBlur}(x_{\text{adv}})$ 
9:     JPEG Compression:  $x_{\text{def}} \leftarrow \text{JPEGCompress}(x_{\text{adv}})$ 
10:    Median Filter:  $x_{\text{def}} \leftarrow \text{MedianFilter}(x_{\text{adv}})$ 
11:    Predict:  $\hat{y} \leftarrow f_\theta(x_{\text{def}})$ 
12:    Compute accuracy:  $\text{Acc} \leftarrow \text{Accuracy}(\hat{y}, y)$ 
13:    Print  $\epsilon$ , Acc
14: end for

```

This algorithm evaluates different defense mechanisms (Gaussian Blur, JPEG Compression, Median Filter) against FGSM adversarial attacks. For each perturbation strength ϵ , it generates adversarial examples, applies a defense layer, and measures the model's accuracy, providing a comprehensive comparison of defense effectiveness.

4.4 Training with Multiple Epsilons

In addition to input-based defenses, it is possible to enhance the model's inherent robustness by incorporating adversarial examples during training. That involves generating adversarial examples from the training dataset and including them in the training process. This helps the model learn more resilient decision boundaries.

Rather than using a single, fixed perturbation strength ϵ , adversarial samples were generated using multiple epsilon values: $\epsilon \in \{0.01, 0.05, 0.1, 0.2, 0.3\}$.

This approach led to a substantial improvement in the model's robustness to adversarial attacks, particularly in the lower to mid-range epsilon values (e.g., 0.01–0.1). Compared to the baseline model, the accuracy showed less significant degradation for higher epsilon values.

Algorithm 4 Adversarial Training with Multiple Epsilon Values

- 1: **Input:** Clean training data (x, y) , model f_θ , list of ϵ values
 - 2: **Output:** Trained robust model parameters θ
 - 3: Initialize adversarial dataset $\mathcal{D}_{adv} \leftarrow \emptyset$
 - 4: **for** each ϵ in $\{\epsilon_1, \epsilon_2, \dots, \epsilon_n\}$ **do**
 - 5: **for** each (x_i, y_i) in training set **do**
 - 6: Compute gradient: $g \leftarrow \nabla_x J(f_\theta(x_i), y_i)$
 - 7: Generate adversarial example: $x_{adv} \leftarrow x_i + \epsilon \cdot \text{sign}(g)$
 - 8: Add (x_{adv}, y_i) to $\mathcal{D}_{adv}$
 - 9: **end for**
 - 10: **end for**
 - 11: Combine datasets: $\mathcal{D}_{mix} \leftarrow \mathcal{D}_{clean} \cup \mathcal{D}_{adv}$
 - 12: Train model f_θ on $\mathcal{D}_{mix}$ using cross-entropy loss
-

This algorithm enhances model robustness by generating adversarial examples at multiple perturbation strengths (ϵ values). It creates a diverse adversarial training set by attacking the model with different attack magnitudes, then combines these with clean data to train a more comprehensive defense. The multi- ϵ approach helps the model learn to resist attacks of varying intensities.

Chapter 5

Results Analysis

Our proposed scheme undergoes comprehensive simulated evaluation in this section across different performance evaluation scenarios. Our evaluation entails a comparative study between our proposed scheme and two alternative approaches.

5.1 Models Description

This thesis utilizes four distinct deep learning models across various experiments to evaluate and defend against FGSM adversarial attacks. The first model, ResNet34, was used to use the FGSM attack and use adversarial training as a defense on a Kaggle dataset. ResNet34 is a robust 34-layer pretrained convolutional neural network, adapted for the particular classification task by replacing its last layer. With about 21.8 million trainable parameters, it was a good baseline to evaluate the effect of the FGSM attacks and the efficacy of adversarial training on a real-world dataset.

The second model, known as *myCNN*, is a manually designed, slim CNN. It has two convolutional layers and two fully connected layers with dropout, totaling roughly 1.2 million parameters. This model was utilized to study the influence of FGSM attacks and examine performance enhancements through adversarial training on the MNIST dataset.

The third model, titled *DefendedModel*, incorporates a defense mechanism—like smoothing or noise-reduction filters—before the CNN layers. This preprocessing layer was implemented to counter adversarial perturbations, and its performance was examined under different values of epsilon.

Lastly, the *NoisyModel* kept the architecture of *myCNN*, but added Gaussian

noise to the training data as a way of image augmentation. This was with the aim of increasing the robustness of the model by allowing it to learn more generalized features so that it becomes less vulnerable to small adversarial perturbations.

All the models were trained on the Cross-Entropy Loss function and tested on their capability against FGSM attacks, with and without employing the respective defense mechanisms. The models used Cross-Entropy Loss for training and underwent testing against FGSM attacks by using or not using defense mechanisms.

5.2 Dataset Description

The evaluation of adversarial attacks and defense mechanisms used two main datasets. The highly evaluated MNIST dataset serves as the test platform for handwritten digits from 0 through 9. The MNIST dataset presents grayscale images which are 28×28 pixels large and contain 70,000 pictures between training and testing sets split into 60,000 and 10,000 respectively using ten different categories for classification. The basics of both adversarial attack techniques and lightweight defense mechanisms align perfectly with the characteristics of the MNIST dataset because it possesses simple standardized structure.

The second dataset is an image dataset of natural scenes from Kaggle, representing a real-world image classification challenge. It contains 25,000 images spread across 6 classes. A Kaggle dataset served to assess how deep pretrained ResNet50 operated during both FGSM attacks and adversarial training procedures. It provided valuable insights into how more complex, real-world data behaves under adversarial conditions.

Together, these datasets enabled a comprehensive evaluation across both simple and complex domains, ensuring a robust assessment of the proposed defense techniques.

5.3 Adversarial Training Analysis

To evaluate the vulnerability of the ResNet-34 model to adversarial attacks, we applied the FGSM with varying values of the perturbation parameter ϵ . The results are documented in terms of model accuracy before the attack, during the attack for specific ϵ values, and after adversarial training.

	precision	recall	f1-score	support
buildings	0.92	0.92	0.92	349
forest	0.99	0.99	0.99	379
glacier	0.88	0.88	0.88	442
mountain	0.89	0.87	0.88	420
sea	0.93	0.96	0.95	408
street	0.93	0.94	0.93	400
accuracy			0.92	2398
macro avg	0.92	0.93	0.93	2398
weighted avg	0.92	0.92	0.92	2398

Figure 5.1: Accuracy of ResNet-34 before FGSM attack

```
For Epsilon =0.25
/usr/local/lib/python3.10/dist-packages/torch/utils/data/
warnings.warn(_create_warning_msg(
```

	precision	recall	f1-score	support
buildings	0.43	0.68	0.53	349
forest	0.72	0.47	0.57	379
glacier	0.28	0.33	0.30	442
mountain	0.48	0.31	0.38	420
sea	0.29	0.49	0.36	408
street	0.36	0.09	0.15	400
accuracy			0.39	2398
macro avg	0.43	0.40	0.38	2398
weighted avg	0.42	0.39	0.38	2398

Figure 5.2: Accuracy on $\epsilon = 0.25$.

Initially, the ResNet-34 model reached to an accuracy of **92%** on the clean dataset, which serves as a performance benchmark before any adversarial perturbations are introduced. Subsequently, we performed FGSM attacks with two representative epsilon

```
For Epsilon =0.3
/usr/local/lib/python3.10/dist-packages/torch/utils/data/
warnings.warn(_create_warning_msg(
```

	precision	recall	f1-score	support
buildings	0.43	0.66	0.52	349
forest	0.65	0.45	0.54	379
glacier	0.28	0.32	0.30	442
mountain	0.49	0.30	0.37	420
sea	0.28	0.51	0.36	408
street	0.32	0.08	0.12	400
accuracy			0.38	2398
macro avg	0.41	0.39	0.37	2398
weighted avg	0.40	0.38	0.36	2398

Figure 5.3: Accuracy on $\epsilon = 0.3$.

values, $\epsilon = 0.25$ and $\epsilon = 0.30$, to observe how performance degrades with increasing perturbation strength. The accuracy dropped to **39%** and **38%** respectively.

These results clearly demonstrate the susceptibility of the model to even relatively small adversarial perturbations.

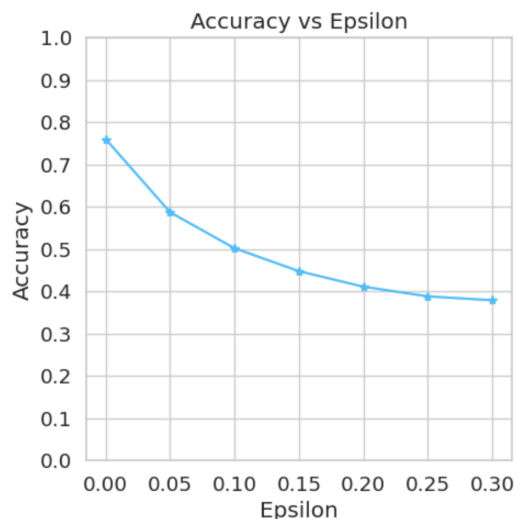


Figure 5.4: Accuracy vs Epsilon.

A more detailed analysis is shown in Figure 5.4, where we plot the model’s classification accuracy as a function of different epsilon values ranging from low to high. The graph exhibits a clear inverse relationship between epsilon and accuracy, consistent with the theoretical understanding of FGSM: as the perturbation increases, the adversarial examples deviate further from the original data distribution, making it harder for the model to classify them correctly. To improve the model’s robustness, we

	precision	recall	f1-score	support
buildings	0.76	0.84	0.80	349
forest	0.95	0.89	0.92	379
glacier	0.84	0.70	0.76	442
mountain	0.82	0.72	0.76	420
sea	0.70	0.89	0.79	408
street	0.82	0.82	0.82	400
accuracy			0.81	2398
macro avg	0.81	0.81	0.81	2398
weighted avg	0.81	0.81	0.81	2398

Figure 5.5: Accuracy After Adversarial Training.

retrained it on adversarial examples. This adversarial training significantly enhanced

the model’s resistance to attacks. After retraining, the model achieved **81.3%** under the same FGSM conditions, showing a substantial recovery in performance.

These findings confirm that while standard CNNs like ResNet-34 are vulnerable to gradient-based adversarial attacks, integrating adversarial training techniques can mitigate such vulnerabilities and produce more robust models.

5.4 Noise Injection Analysis

To investigate the effectiveness of noise-based augmentation techniques as a defense against adversarial attacks, multiple noise injection methods were applied during training. These techniques aimed to improve the robustness of the model by forcing it to learn robust and generalized features. The primary techniques used include Random Gaussian Noise (RGN), which introduces pixel-level fluctuations following a normal distribution; Stochastic Motion Blur, which simulates camera shake or object movement to challenge the model with real-world image degradation; Stochastic Glass Blur (SGB), which distorts local image regions by applying localized blurring and pixel shuffling; and Random Sized Coarse Dropout (RSCD), which randomly removes rectangular patches from the image, compelling the model to learn discriminative features from incomplete visual information. Among these, RGN and RSCD showed notable improvements in adversarial robustness, while SGB and Motion Blur contributed to better generalization under distorted inputs.

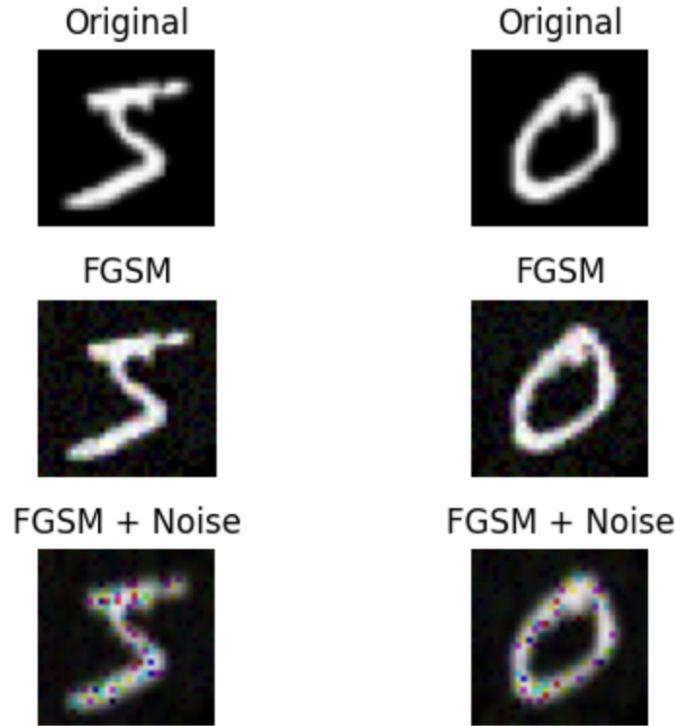


Figure 5.6: Images After FGSM And Noise Injection.

Figure 5.7 illustrate the the original images , image after the FGSM attack and images after the noise injection. After Noise injection there are some dotted circle on the images.

The accuracy results presented in the Figure 5.7 illustrate the comparative performance of these techniques, confirming that noise injection not only acts as a regularizer but also strengthens the model's ability to withstand FGSM adversarial perturbations.

```
Clean Test Accuracy: 92.40%
FGSM + Fixed Noise Accuracy: 74.40%
```

Figure 5.7: Accuracy After Noise Injection.

5.5 Defense Layer Analysis

The parts of thesis determines how well the implemented defense methods perform after facing FGSM adversarial attacks, we analyzed the performance of the DefendedModel, a CNN architecture enhanced with an initial defense layer. This layer is responsible for preprocessing inputs before classification, aiming to reduce the influence of adversarial perturbations.

```
Epoch 20/20
422/422 ----- 5s 9ms/step - accuracy: 0.9982
313/313 ----- 2s 4ms/step - accuracy: 0.9911
Accuracy on test data is : 99.19999837875366
4/4 ----- 1s 215ms/step
Accuracy on perturbed images: 67.00%
```

Figure 5.8: DefendedModel Accuracy.

Figure 5.8 tells the classification accuracy of the DefendedModel on adversarial examples. The model demonstrates significantly higher accuracy than the baseline, indicating that the added defense layer effectively mitigates the impact of FGSM perturbations.

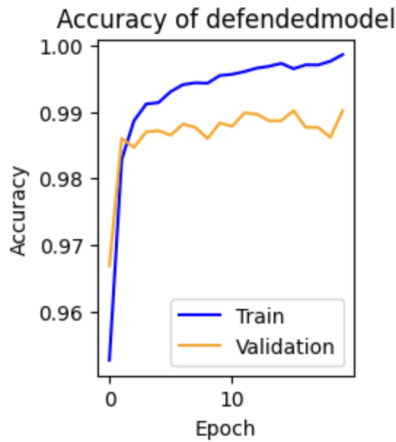


Figure 5.9: Accuracy vs Epoch.

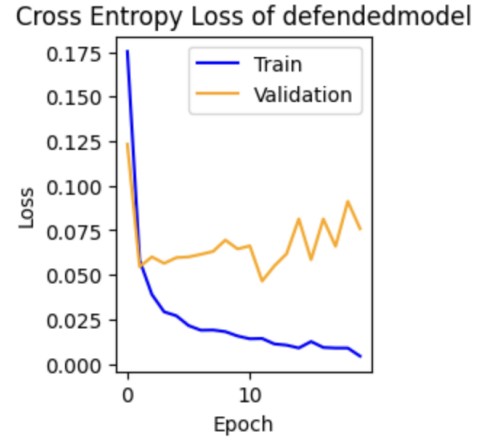


Figure 5.10: Loss vs Epoch.

The training and validation trends of the DefendedModel are illustrated in Figure 5.9 and Figure 5.10, which display the model's accuracy and loss over training epochs. The smooth convergence and stable accuracy curves confirm that the addition of the defense layer does not hinder the model's learning capability but rather promotes better generalization, even under attack scenarios.

In addition to the general defense layer used in the DefendedModel, we also evaluated the impact of three specific image preprocessing techniques as standalone defenses:

- **Gaussian Blur Analysis:** This method applies a smoothing effect that reduces local variations in the input image. By blurring high-frequency adversarial noise, the classifier is exposed to a more stable input. The performance using Gaussian blur is shown in Figure 5.11.

```

4/4 ————— 0s 8ms/step
Epsilon: 0.000 => Accuracy after Gaussian blur: 98.00%
4/4 ————— 0s 8ms/step
Epsilon: 0.007 => Accuracy after Gaussian blur: 98.00%
4/4 ————— 0s 9ms/step
Epsilon: 0.010 => Accuracy after Gaussian blur: 98.00%
4/4 ————— 0s 8ms/step
Epsilon: 0.020 => Accuracy after Gaussian blur: 97.00%
4/4 ————— 0s 11ms/step
Epsilon: 0.030 => Accuracy after Gaussian blur: 97.00%
4/4 ————— 0s 8ms/step
Epsilon: 0.050 => Accuracy after Gaussian blur: 93.00%
4/4 ————— 0s 8ms/step
Epsilon: 0.100 => Accuracy after Gaussian blur: 85.00%
4/4 ————— 0s 13ms/step
Epsilon: 0.200 => Accuracy after Gaussian blur: 72.00%
4/4 ————— 0s 8ms/step
Epsilon: 0.300 => Accuracy after Gaussian blur: 52.00%
```

Figure 5.11: GaussianBlur Defense Layer.

- **JPEG Compression Analysis:** This technique introduces controlled quantization and decompression, effectively removing subtle perturbations introduced by adversarial attacks. It acts as a noise filter by reducing low-amplitude changes. The corresponding accuracy image is displayed in Figure 5.12.

```

--- JPEG Compression Defense ---
4/4 ----- 0s 8ms/step
Epsilon: 0.000 => Accuracy after JPEG compression: 99.00%
4/4 ----- 0s 9ms/step
Epsilon: 0.007 => Accuracy after JPEG compression: 99.00%
4/4 ----- 0s 8ms/step
Epsilon: 0.010 => Accuracy after JPEG compression: 99.00%
4/4 ----- 0s 9ms/step
Epsilon: 0.020 => Accuracy after JPEG compression: 99.00%
4/4 ----- 0s 8ms/step
Epsilon: 0.030 => Accuracy after JPEG compression: 98.00%
4/4 ----- 0s 8ms/step
Epsilon: 0.050 => Accuracy after JPEG compression: 94.00%
4/4 ----- 0s 8ms/step
Epsilon: 0.100 => Accuracy after JPEG compression: 79.00%
4/4 ----- 0s 8ms/step
Epsilon: 0.200 => Accuracy after JPEG compression: 63.00%
4/4 ----- 0s 8ms/step
Epsilon: 0.300 => Accuracy after JPEG compression: 31.00%

```

Figure 5.12: JPEGCompression Defense Layer.

- **Median Denoising Analysis:** This method replaces each pixel's value with the median of neighboring pixels, effectively eliminating impulse noise or adversarial artifacts. Figure 5.13 shows the resulting model accuracy with this technique applied.

```

--- Median Denoising Defense ---
4/4 ----- 0s 8ms/step
Epsilon: 0.000 => Accuracy after Median Denoising: 100.00%
4/4 ----- 0s 11ms/step
Epsilon: 0.007 => Accuracy after Median Denoising: 100.00%
4/4 ----- 0s 8ms/step
Epsilon: 0.010 => Accuracy after Median Denoising: 100.00%
4/4 ----- 0s 8ms/step
Epsilon: 0.020 => Accuracy after Median Denoising: 98.00%
4/4 ----- 0s 12ms/step
Epsilon: 0.030 => Accuracy after Median Denoising: 97.00%
4/4 ----- 0s 8ms/step
Epsilon: 0.050 => Accuracy after Median Denoising: 92.00%
4/4 ----- 0s 8ms/step
Epsilon: 0.100 => Accuracy after Median Denoising: 81.00%
4/4 ----- 0s 11ms/step
Epsilon: 0.200 => Accuracy after Median Denoising: 57.00%
4/4 ----- 0s 8ms/step
Epsilon: 0.300 => Accuracy after Median Denoising: 32.00%

```

Figure 5.13: Median Denoising Defense Layer.

Each of these preprocessing techniques shows varying levels of effectiveness in defending against adversarial inputs. Among them, Gaussian blur and JPEG compression show more consistent improvements, while median denoising helps in filtering out specific perturbations effectively. These results confirm that layer-based prepro-

cessing can act as a lightweight, non-learnable defense that can be easily integrated with existing models.

5.6 Epsilon Training Analysis

To evaluate the effectiveness of adversarial training in enhancing model robustness, we present the accuracy of the model before and after the adversarial training process. Different values of epsilon make the model more robust as it adds the diversity. Bigger value of epsilon means the adversarial images are more far from the decision boundary.

```

4/4 ----- 0s 8ms/step
Epsilon: 0.000 => Accuracy on perturbed images: 99.00%
4/4 ----- 0s 8ms/step
Epsilon: 0.007 => Accuracy on perturbed images: 99.00%
4/4 ----- 0s 12ms/step
Epsilon: 0.010 => Accuracy on perturbed images: 99.00%
4/4 ----- 0s 8ms/step
Epsilon: 0.020 => Accuracy on perturbed images: 98.00%
4/4 ----- 0s 9ms/step
Epsilon: 0.030 => Accuracy on perturbed images: 97.00%
4/4 ----- 0s 13ms/step
Epsilon: 0.050 => Accuracy on perturbed images: 90.00%
4/4 ----- 0s 8ms/step
Epsilon: 0.100 => Accuracy on perturbed images: 78.00%
4/4 ----- 0s 8ms/step
Epsilon: 0.200 => Accuracy on perturbed images: 59.00%
4/4 ----- 0s 8ms/step
Epsilon: 0.300 => Accuracy on perturbed images: 31.00%

```

Figure 5.14: Accuracy On Various Epsilon.

Figure 5.14 shows the model's accuracy before to epsilon training, highlighting the performance on both clean and adversarially perturbed inputs. This provides a baseline for understanding the weakness of the model against adversarial(FGSM) attacks.

```

--- Accuracy on FGSM Adversarial Examples (Post-Adversarial Train.
4/4 ----- 0s 8ms/step
Epsilon: 0.010 => Accuracy after adversarial training: 99.00%
4/4 ----- 0s 9ms/step
Epsilon: 0.050 => Accuracy after adversarial training: 90.00%
4/4 ----- 0s 8ms/step
Epsilon: 0.100 => Accuracy after adversarial training: 84.00%
4/4 ----- 0s 8ms/step
Epsilon: 0.200 => Accuracy after adversarial training: 73.00%
4/4 ----- 0s 8ms/step
Epsilon: 0.300 => Accuracy after adversarial training: 56.00%

```

Figure 5.15: After Epsilon Training.

In Figure 5.15 , we present the accuracy of the model after epsilon adversarial training. As seen, the model demonstrates a marked improvement in its ability to classify adversarial examples correctly. The adversarial training process implementing adversarial examples exposure during model training leads to better accuracies which demonstrates successful model defense against FGSM attack perturbations.

These results underscore the significance of epsilon training in improving model resilience and ensuring reliable performance in adversarial settings.

Chapter 6

Conclusion and Future Scope

This thesis explored the vulnerability of deep learning models to adversarial attacks, namely FGSM, and tested various defense approaches that are intended to increase model robustness. Experimentation was carried out on two data sets: the Kaggle data set and the traditional MNIST data set. A sophisticated neural network, ResNet-34, was utilized to process the Kaggle data set, whereas a self-created myCNN was used for MNIST. It was discovered that adversarial training significantly enhanced the robustness of the model by retraining the model on perturbed data. Moreover, defense layers were added to preprocess inputs and remove adversarial noise before classification. The model, named the DefendedModel, which has such a preprocessing layer, performed better at different epsilon values.

Another approach investigated was noise injection during training, using methods like Gaussian noise, stochastic motion blur, glass blur, and coarse dropout. These approaches assisted the model in learning more generalized features, leading to increased resistance against adversarial inputs. In general, this thesis proved that the combination of adversarial training, defensive architecture, and data augmentation methods can effectively enhance robustness while keeping relatively high clean accuracy.

The next step for future research is in a few areas. More advanced and stronger attacks, including PGD, DeepFool, and Carlini-Wagner, may be investigated to check the efficiency of existing defense schemes under harsher conditions. More importantly, design of adaptive or learnable defensive schemes that have the capability of responding to varied perturbations dynamically is a promising avenue to pursue further. Certified defenses that make mathematical guarantees, e.g., randomized smoothing or robust training with certification bounds, are of enormous promise for

future research. Also, experimenting on these defense strategies with larger and more diverse sets, e.g., CIFAR-10, CIFAR-100, or domain-specific datasets (e.g., medical images), would provide information about scalability and real-world applicability of these techniques. Lastly, deploying these methods in federated learning or edge computing settings may provide solutions to practical security issues in distributed and resource-scarce environments. This work lays a good foundation for reliable AI and ML systems that can withstand adversarial attacks.

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Chapter 1

Introduction

1.1 Background

Deep learning has significantly impacted various sectors including autonomous vehicles, healthcare systems, biometric identification, and surveillance technologies. Among the most effective tools in this space are Convolutional Neural Networks (CNNs), which have demonstrated high accuracy and strong generalization capabilities in handling complex image-based tasks. Nevertheless, these models are highly susceptible to adversarial attacks—small, intentional perturbations introduced to input data to mislead the model, often without any noticeable change to the human eye. One of the earliest forms of adversarial attack appears through the implementation of Fast Gradient Sign Method (FGSM). [7]. The procedure determines minor data modifications according to loss function gradients which makes the network mistake its data classification decisions. The simplicity and effectiveness of FGSM highlight the inherent weaknesses of deep learning systems, especially when applied in sensitive areas like medical imaging, facial authentication, or self-driving technology, where prediction errors can have serious implications.

As adversarial machine learning has matured, research has expanded to include both advanced attack strategies and a variety of defense mechanisms. Defensive measures include methods like adversarial training—where models are retrained using adversarial inputs—and input preprocessing techniques such as smoothing, denoising, or compression. However, many of these defenses face limitations: they may reduce accuracy on unperturbed inputs, require substantial computational power, or lack effectiveness against a broad spectrum of attacks.

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